Sharp interface dynamics for a two-phase model of compressible fluids

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Presentation of the model

This poster presents a qualitative analysis of a model that describes the motion of two compressible immiscible fluids. Such flows arise in various contexts, to name a few examples:

- ☐ **biological systems**, such as air–liquid flows in the respiratory tract;
- ☐ **geophysical phenomena**, such as ice formation, river flooding, landslides, snow avalanches;
- ☐ **industrial applications**, such as bubbly flows in aeration columns and oil–water flows in pipelines.

Each phase of the flow may obey its own constitutive laws (e.g., pressure and viscosity laws). Since the phases are assumed to be immiscible, they are separated by a zero-thickness interface (a sharp interface), across which flow variables such as density and temperature may be discontinuous. A natural question concerning the dynamics of this sharp interface is: *does an initially regular interface remain regular over time?*

The dynamics of the mixture can be described by the volume fraction c of one phase, the total mixture density $\rho = \rho(t,x)$, and the common velocity field $u(t,x) \in \mathbb{R}^d$. These quantities satisfy:

$$\begin{cases} \partial_t c + u \cdot \nabla c = 0, \\ \partial_t \rho + \operatorname{div}(\rho u) = 0, \\ \partial_t(\rho u) + \operatorname{div}(\rho u \otimes u) + \nabla P(\rho, c) = \operatorname{div}\big(2\mu(\rho,c)\mathbb{D}u\big) + \nabla\big(\lambda(\rho,c)\operatorname{div} u\big), \\ (c, \rho, \rho u)_{|t=0} = (c_0, \rho_0, \rho_0 u_0). \end{cases} \tag{1}$$

Main problem. For initially immiscible phases, we have $c_0 = \mathbb{1}_{D_0}$ for an open domain D_0; in particular, $c_0(1-c_0) = 0$. Based on DiPerna–Lions theory, any weak solution (c,ρ,u) to (1) in the energy class (see [Vasseur et al., 2019, Novotný, 2020]) with bounded $\operatorname{div} u$ preserves the patch structure of the volume fraction: $c(t)(1-c(t)) = 0$, and hence the immiscibility of the flow. However, the velocity field is only known to be square integrable, which is insufficient to guarantee a well-defined classical flow map for tracking the evolution of the sharp interface.
Our aim is therefore to construct solutions in an intermediate regularity class that allows for discontinuous density while ensuring sufficient regularity of the velocity field to track the evolution of the free boundary $\partial D(t)$ and propagate its regularity over time.

An example of two immiscible phases



Fig. 1: Air bubbles rising in water: the gas and liquid are separated by sharp interfaces.

Main result

Theorem 1 ([Zodji, 2025])**.**
Let $0 < \alpha < 1$ and let $D_0 \subset \mathbb{R}^3$ be an open, bounded, simply connected domain of class $\mathscr{C}^{1+\alpha}$.
Assume that

$$c_0 = \mathbb{1}_{D_0}, \qquad 0 < \rho_* \leqslant \rho_0 \leqslant \rho^*, \qquad \rho_0 \in \mathscr{C}^\alpha(\overline{D_0}) \cap \mathscr{C}^\alpha(\mathbb{R}^3 \setminus D_0).$$

Let $\widetilde{P}$ and $\widetilde{\mu}$ denote the equilibrium pressure and shear viscosity, respectively. Let $q \in (1, 6/5)$ and $u_0 \in H^1(\mathbb{R}^3)$. If $\alpha \in [1/2, 1)$, we further assume that

$$(\rho \dot{u})_{|t=0} = \operatorname{div}\left[2\mu(\rho_0,c_0)\mathbb{D}u_0 + \big(\lambda(\rho_0,c_0)\operatorname{div} u_0 - P(\rho_0,c_0)\big)\mathbb{I}_3\right] \in \dot{H}^{-\check{\alpha}}(\mathbb{R}^3), \tag{2}$$

for some $0 < \check{\alpha} < 3/2 - \alpha$.
There exists $\varepsilon > 0$ such that if

$$\|P(\rho_0,c_0) - \widetilde{P},\, u_0\|^2_{L^2 \cap L^q(\mathbb{R}^3)} + \|\nabla u_0\|^2_{L^2(\mathbb{R}^3)} + \|\mu(\rho_0,c_0) - \widetilde{\mu}\|^2_{L^\infty(\mathbb{R}^3)} \leqslant \varepsilon, \tag{3}$$

then there exists a unique global weak solution (c,ρ,u) to (1) with $\nabla u \in L^1((0,\infty); L^\infty(\mathbb{R}^3))$. Moreover, for every $t > 0$:

$$c(t) = \mathbb{1}_{D(t)}, \qquad \partial D(t) \in \mathscr{C}^{1+\alpha}, \qquad \rho(t) \in \mathscr{C}^\alpha(\overline{D(t)}) \cap C^\alpha(\mathbb{R}^3 \setminus D(t)),$$

where $D(t)$ is the image of D_0 under the flow generated by the velocity field u.

Remark 2

- ☐ The smallness condition on the shear viscosity fluctuation in (3) can be replaced by

  $$\|[\![\mu(\rho_0,c_0),\, P(\rho_0,c_0)]\!]\|_{L^\infty(\partial D_0)} \ll 1,$$

  provided that $\mu(\rho,c) = \mu(c)$ and that there exists a function $\Psi \in W^{2,\infty}_{\mathrm{loc}}$ such that $P = \Psi(f)$.
  Here $[\![\cdot]\!]$ denotes the jump across the interface, and f satisfies $\rho\partial_\rho f = 2\mu + \lambda$. This condition holds, for example, in the one-phase setting or when pressure and total viscosity $2\mu + \lambda$ are proportional.
- ☐ Under the assumptions of the previous item, if there exists a function $\Psi_\mu \in W^{2,\infty}_{\mathrm{loc}}$ such that $\mu = \Psi_\mu(f)$, then the shear-viscosity and the pressure jumps decay exponentially in time:

  $$\|[\![f(\rho,c)(t)]\!]\|_{L^\infty(\partial D(t))} \lesssim e^{-c_* t + C_* \int_0^t \|\nabla u\|_{L^\infty}} \|[\![f(\rho_0,c_0)]\!]\|_{L^\infty(\partial D_0)}. \tag{4}$$

Main tools

The key step in the proof is to establish sufficient regularity of the velocity field, despite the discontinuity of the density, to ensure that the free boundary remains regular over time. In our setting, this amounts to deriving piecewise $\mathcal{C}^\alpha$ estimates for the velocity gradient.

- ☐ **Expression of the velocity gradient.** From the momentum equation, we derive the following expression for the velocity gradient:

  $$\begin{aligned} \nabla u = &-\frac{1}{\mu(\rho,c)}\nabla(-\Delta)^{-1}\mathbb{P}(\rho\dot{u}) + \nabla(-\Delta)^{-1}\nabla\left(\frac{1}{\nu(\rho,c)}(-\Delta)^{-1}\operatorname{div}(\rho\dot{u})\right) - \nabla(-\Delta)^{-1}\nabla\left(\frac{P(\rho,c)-\widetilde{P}}{\nu(\rho,c)}\right) \\ &+ \frac{1}{\mu(\rho,c)}\big[\mathcal{K}, \mu(\rho,c) - \widetilde{\mu}\big](2\mathbb{D}u) - \nabla(-\Delta)^{-1}\nabla\left(\frac{1}{\nu(\rho,c)}\big[\check{\mathcal{K}}, \mu(\rho,c) - \widetilde{\mu}\big](2\mathbb{D}u)\right). \end{aligned} \tag{5}$$

  Here, $\dot{u} = \partial_t u + (u \cdot \nabla)u$ is the material acceleration, $\nu = 2\mu + \lambda$ is the total viscosity coefficient, $\mathbb{P}$ is the Leray projector, and $\mathcal{K}$ and $\check{\mathcal{K}}$ are second-order Riesz operators.
- ☐ **Energy estimates for $\dot{u}$.** We differentiate the momentum equation along the fluid particles' trajectories (using the conservative operator $\partial_t \cdot + \operatorname{div}(u\,\cdot)$) and derive energy estimates for $\dot{u}$. Under the smallness assumption

  $$\sup_{t>0} \|[\![\mu(\rho,c)(t)]\!]\|_{L^\infty(\partial D(t))} \ll 1, \quad \text{these estimates yield} \quad \dot{u} \in L^1((0,\infty); L^{3/(1-\alpha)}(\mathbb{R}^3)).$$

  In particular, this controls the $\mathcal{C}^\alpha$-norm of the first term in (5), apart from the factor $1/\mu(\rho,c)$.
- ☐ **Piecewise Hölder estimates.** The remaining terms in (5) involve Riesz operators and their commutators acting on quantities that may be discontinuous across the interface. In general, Riesz operators are not bounded on $L^\infty(\mathbb{R}^3)$. Building on [Gancedo and García-Juárez, 2023, Hoff, 2002], we establish that even Riesz operators $\mathcal{R}$ act continuously on the space of piecewise Hölder continuous functions across an interface ∂D of class $\mathscr{C}^{1+\alpha}$, with the following bounds for commutators:

  $$\|[\mathcal{R}, w]g\|_{L^\infty(\mathbb{R}^d)} \lesssim \|[\![w]\!]\|_{L^\infty(\partial D)} \|g\|_{\mathscr{C}^\alpha(\overline{D}) \cap \mathscr{C}^\alpha(\mathbb{R}^d \setminus D)} + \text{l.o.t. in } g, \tag{6}$$

  $$|[\mathcal{R}, w]g|_{\dot{\mathscr{C}}^\alpha(\overline{D}) \cap \dot{\mathscr{C}}^\alpha(\mathbb{R}^d \setminus D)} \lesssim \|[\![w]\!]\|_{L^\infty(\partial D)} |g|_{\dot{\mathscr{C}}^\alpha(\overline{D}) \cap \dot{\mathscr{C}}^\alpha(\mathbb{R}^d \setminus D)} + \text{l.o.t. in } g. \tag{7}$$

  We apply these estimates to the remaining terms in (5) and use the smallness of the shear viscosity jump to absorb the terms involving the piecewise $\mathscr{C}^\alpha$-norm of ∇u.
- ☐ **Piecewise regularity of the density.** From the continuity and momentum equations, we deduce

  $$\partial_t f(\rho,c) + u \cdot \nabla f(\rho,c) + P(\rho,c) - \widetilde{P} = (-\Delta)^{-1}\operatorname{div}(\rho\dot{u}) - [\check{\mathcal{K}}, \mu(\rho,c) - \widetilde{\mu}]\mathbb{D}u, \tag{8}$$

  where $f = f(\rho,c)$ satisfies $\rho\partial_\rho f(\rho,c) = \nu(\rho,c)$. The pressure term provides damping, while the piecewise Hölder regularity of the right-hand side follows from the previous two estimates. Together, these properties help propagate the piecewise regularity of the density.
- ☐ **Propagation of the viscosity jump.** To close the estimates, we need to propagate the smallness of the shear viscosity jump.

  Let ψ denote the flow map of the fluid velocity u, and set $\varrho = \rho \circ \psi$. In the setting $P = \Psi(f)$ and $\mu = \Psi_\mu(f)$, taking the jump of (8) across the interface and evaluating it along the flow yields

  $$\frac{d}{dt}[\![f(\varrho,c_0)]\!] + \frac{[\![P(\varrho,c_0)]\!]}{[\![f(\varrho,c_0)]\!]}[\![f(\varrho,c_0)]\!] = \frac{[\![\mu(\varrho,c_0)]\!]}{[\![f(\varrho,c_0)]\!]}[\![f(\varrho,c_0)]\!]\mathcal{O}(\nabla u) \circ \psi.$$

  The regularity of μ and P, together with the strict monotonicity of P, yields the following bounds uniformly on ∂D_0:

  $$\inf_{t \geqslant 0}\left(\frac{[\![P(\varrho(t),c_0)]\!]}{[\![f(\varrho(t),c_0)]\!]}\right) \geqslant c_* > 0, \qquad \sup_{t \geqslant 0}\left|\frac{[\![\mu(\varrho(t),c_0)]\!]}{[\![f(\varrho(t),c_0)]\!]}\right| \leqslant C_*.$$

  These bounds yield

  $$\|[\![f(t)]\!]\|_{L^\infty(\partial D(t))} \lesssim e^{-c_* t + C_* \int_0^t \|\nabla u(s)\|_{L^\infty(\mathbb{R}^3)}\, ds} \|[\![f(0)]\!]\|_{L^\infty(\partial D_0)}.$$

References

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